

Literature-implied status for Oikhberg–Spinu’s Schur-lattice remark

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Status. This is a *literature-implied answer*, not a new proof. In Oikhberg–Spinu, *Domination of operators in the non-commutative setting*, arXiv:1209.0699, Remark **rem:not 1.1** in the Positive Schur Property subsection says that an order-continuous atomic Banach lattice with the Schur property need not be isomorphic to ℓ_1 : if (E_n) are finite-dimensional lattices, then

$$E = \left(\sum_{n=1}^{\infty} E_n \right)_{\ell_1}$$

has the Schur property, and the choice $E_n = \ell_2^n$ is not isomorphic to ℓ_1 . The remark then says that the authors do not know any Banach lattice with the Schur property which is not isomorphic to an ℓ_1 -sum of finite-dimensional spaces.

The supporting example. Kacena–Kalenda–Spurny, *Quantitative Dunford-Pettis property*, arXiv:1110.1243, Example 10.1, constructs a Banach space X such that X^* is a separable L -embedded space with the Schur property. In the proof, for each $\alpha > 0$ they define X_α on c_0 and compute

$$\|x^*\|_\alpha^* = \alpha \|x^*\|_1 + \|x^*\|_\infty \quad (x^* \in \ell_1).$$

Then

$$X = \left(\bigoplus_{n \in \mathbb{N}} X_{1/n} \right)_{c_0}, \quad X^* = \left(\bigoplus_{n \in \mathbb{N}} X_{1/n}^* \right)_{\ell_1},$$

and the paper states that this X^* has the Schur property.

Each X_α^* is a Banach lattice under the usual coordinate order on ℓ_1 : if $|u| \leq |v|$, then both $\|u\|_1 \leq \|v\|_1$ and $\|u\|_\infty \leq \|v\|_\infty$. The norm is equivalent to the usual ℓ_1 -norm, so the space is complete; it is atomic and order-continuous by the usual monotone-convergence argument on ℓ_1 . Therefore

$$E := X^* = \left(\bigoplus_{n \in \mathbb{N}} X_{1/n}^* \right)_{\ell_1}$$

is an atomic order-continuous Banach lattice with the Schur property.

Why it is not an ℓ_1 -sum of finite-dimensional spaces. Kalenda, *Quantitative Schur property and measures of weak non-compactness*, arXiv:2505.12893, defines the c -Schur property by

$$\text{ca}(x_n) \leq c \delta(x_n)$$

for every bounded sequence (x_n) , where $\delta(x_n)$ is the usual measure of weak non-Cauchyness, equivalently the diameter of the set of weak-star cluster points of (x_n) in the bidual. The same paper

notes, in Example (2) after Corollary `cor:qschur`, that if X is the space from Kacena–Kalenda–Spurny Example 10.1, then X^* is L -embedded and has the Schur property, but $\text{wk}(\cdot)$ and $\omega(\cdot)$ are not equivalent in X^* ; hence X^* fails the quantitative Schur property.

On the other hand, Kalenda’s Theorem `T:ell1suma` says that, for $c \geq 1$, the c -Schur property is preserved by arbitrary ℓ_1 -sums. Every finite-dimensional Banach space has the 1-Schur property: bounded sequences have compact norm closure, and the diameter of the norm cluster set coincides with $\text{ca}(x_n)$, while the weak and norm cluster sets coincide in finite dimension. Thus every ℓ_1 -sum of finite-dimensional spaces has the quantitative Schur property.

Finally, quantitative Schur is invariant under Banach-space isomorphism up to changing the constant. If $T : Y \rightarrow Z$ is an isomorphism and Y has the c -Schur property, then for $z_n = Ty_n$,

$$\text{ca}(z_n) \leq \|T\| \text{ca}(y_n) \leq c\|T\| \delta_Y(y_n) \leq c\|T\| \|T^{-1}\| \delta_Z(z_n).$$

Consequently any space isomorphic to an ℓ_1 -sum of finite-dimensional spaces must have the quantitative Schur property. Since the above Kacena–Kalenda–Spurny X^* fails quantitative Schur, it is not isomorphic, even as a Banach space, to an ℓ_1 -sum of finite-dimensional spaces.

Conclusion and scope. The space

$$E = \left(\bigoplus_{n \in \mathbb{N}} (\ell_1, \|x\|_{1/n} = (1/n)\|x\|_1 + \|x\|_\infty) \right)_{\ell_1}$$

is a Banach lattice with the Schur property, but it is not isomorphic to an ℓ_1 -sum of finite-dimensional spaces. This supplies a negative answer to the “we do not know of any” remark in arXiv:1209.0699. The supporting papers do not explicitly cite that remark, so the status is `literature_implied_answer` rather than `literature_already_answered`. No claim is made here about the main non-commutative domination questions in the source paper.

References

- [1] T. Oikhberg and E. Spinu, *Domination of operators in the non-commutative setting*, arXiv:1209.0699.
- [2] M. Kacena, O. F. K. Kalenda, and J. Spurny, *Quantitative Dunford-Pettis property*, arXiv:1110.1243.
- [3] O. F. K. Kalenda, *Quantitative Schur property and measures of weak non-compactness*, arXiv:2505.12893.